



Ritha Ch

Azure DevOps/Data Engineer

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Summary:

- ✓ 6 plus years of experience in Executing **DevOps** strategy in various environments of **Linux** and **Windows** servers along with endorsing cloud strategies based on **AWS** and **Azure**.
- ✓ Experience in deploying **Azure IaaS** virtual machines (VMs) and cloud services (**PaaS** role instances) into secure VNets and subnets.
- ✓ Experience in Architecting solutions for Microsoft **Azure** and worked on **Azure web application, App services, Azure storage, Azure SQL Database, Virtual machines, Fabric controller, Azure AD, Azure search, and notification hub**.
- ✓ Experience in Migrating **SQL database** to **Azure data Lake**, migrating **on premise** databases to **Azure Data Lake** store using **Azure Data factory**, migrating infrastructure and application from on premise to Azure.
- ✓ Configured and managed **TCP/IP, DHCP, DNS, and Active Directory**.
- ✓ Created and managed **user accounts, permissions, and access rights**.
- ✓ Strong knowledge of installing, **configuring, tuning, maintaining, and monitoring Windows Server 2008/2012 Enterprise server**.
- ✓ Significantly skillful in Designing and implementing **Microsoft Azure Solutions** across a range of cloud models like **SaaS, PaaS, IaaS**, and integrated **on premise, on-demand workloads** with the **Azure public cloud**.
- ✓ Implemented **ARM Templates** for deployment of the resources, deployed the resources into Azure using the PowerShell and Continuous Integration by **VSTS**.
- ✓ Experienced in writing **Azure Databrick** notebooks in different programming languages like **Pyspark, Scala** and creating pipelines using **Azure Data Factory** to move data from one data lake to other lake, **data lake to data warehouse** and **on-prem to data lake**.
- ✓ Experience on configuring **Azure DevOps** pipeline using **ARM templates** and **PowerShell** to deploy several azure services like **Azure app service, Azure key vault, storage** and used **azure boards, Azure repos** for storing data from multiple resources and created **YAML pipelines** to deploy the resources on to Azure Platform.
- ✓ Deployed **Windows** and **Linux** based **Docker** containers with popular container orchestrators like **Kubernetes** and **DCOS** provisioned by the **Azure Container Service** and integration of container registries, specifically **Docker Hub** and the **Azure Container Registry** into DevOps workflows.

- ✓ Created **Pods** and services using Master and Minion architecture of **Kubernetes** and developed microservices build by integrating **Jenkins** with **Docker** container using **Docker** pipeline plugin, leveraging **Python** allowing for creation and maintenance build jobs and used **Kubernetes** to deploy services.
- ✓ Worked on **Kubernetes** to manage containerized applications using its nodes, **Configuring Maps**, selector services and deployed application containers as **Pods**. To provide **secured network** connectivity to the containers we used **Calico**. Responsible for build and deployment automation using **Docker**, **Kubernetes** containers.
- ✓ We set up **Kubernetes** (k8s) Clusters for running microservices and pushed microservices into production with Kubernetes backed Infrastructure. Development of automation of **Kubernetes clusters** via playbooks in **Ansible**.
- ✓ Provisioning servers/instances using infrastructure as code such as **Terraform** and **CloudFormation**. Code is stored in a private repository and constantly updated.
- ✓ Efficient in writing **Jenkins CI/CD** pipelines to drive all Microservices builds also creating **Groovy** Scripts for setting up Continuous Integration for new branches, build automation, Plugin management, Securing Jenkins.
- ✓ Proficient with **Power Shell**, **JSON**, **YAML**, **Groovy**, and **Power shell** scripting languages.
- ✓ Experienced utilizing Java/J2EE technologies such as **Core Java**, **Java Server Pages (JSP)**, **Servlets**, **JDBC**, **Spring** and **Hibernate**.
- ✓ Experience in developing **end-end test approaches** and managing the end-to-end quality of the applications.
- ✓ Experience in using **JIRA**, **Confluence**, and other agile project management tools.
- ✓ Ability to **communicate**, **collaborate** and work well in a team-oriented environment.
- ✓ Excellent interpersonal, team player and **multi-tasking skills**, participated in daily stand-up meetings, status meeting and retrospective meetings in a distributed team environment.
- ✓ Experience in deploying **Kubernetes Cluster** on **Azure** with master/minion architecture and wrote many **YAML** files to create many services like pods, deployments, auto scaling, load balancers, labels, health checks, Namespaces.
- ✓ Extensively worked on **Chef Roles**, **Cookbooks**, **Recipes**, **Templates**, **Resources**, **Attributes & Data** bags. Proficient in the setup of **Chef Servers**, **workstations & bootstrapping** infrastructure **Chef Nodes** for configuration management.
- ✓ Experience in writing **Jenkins Pipeline Groovy** Scripts for Continuous Integration and build workflows and Used **Jenkins** for uploading **Artifacts** into **Nexus Repository**.
- ✓ Experience in Monitoring server performance with tools like **Nagios**, **Splunk**, **Datadog**, **New Relic** and resolved network related issues with manual commands and built **Splunk Cluster** environment with High Availability resources. implemented **Azure Application Insights** to store user activities and error logging.

- ✓ Experienced utilizing Java/J2EE technologies such as **Core Java**, **Java Server Pages (JSP)**, **Servlets**, **JDBC**, **Spring** and **Hibernate**.

Education:

- ✓ Bachelor's in Electronics and Communications, JNTUK, INDIA Aug'2009-July'2013

Skill Set:

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|---|---|
| ✓ Azure Data Factory, Azure Data Bricks | ✓ Azure Active Directory, Azure Container Service |
| ✓ Azure Kubernetes Service, Azure functions | |
| ✓ Jenkins | ✓ Azure Virtual Machines, RBAC |
| ✓ Java, | ✓ Ansible |
| ✓ JIRA, Azure Boards | ✓ Git and Git Bash |
| ✓ Azure Application Insights, Azure DevOps | ✓ Agile, Waterfall |
| ✓ SQL, Oracle | ✓ Agile |
| ✓ PowerShell, Bash | ✓ AWS |
| ✓ HTML, CSS | ✓ ARM Templates |
| ✓ ARM templates | ✓ Splunk |
| ✓ Scala, Pyspark | ✓ Confluence. |
| ✓ PowerBI | ✓ Terraform |

Professional Experience:

Azure DevOps Engineer

March'2022 – Present

Client: (FTC-FEMA) (Remote).

Responsibilities:

- Created Azure infrastructure using **ARM templates**, **Azure pipelines** to build, test, and release, and managed **Azure Container Registry** to store private **Docker images**, which are deployed. Grafana Monitor to collect metrics and logs.
- Implemented continuous **integration/continuous delivery (CI/CD) pipelines** for deploying applications.
- Azure Virtual machines **maintenance** and fixing **security vulnerabilities**.
- Migrated to **App Service Environment v2** to **App Service Environment v3**.
- Performed automation tasks in **PowerShell**, **Azure CLI** and **JSON** from **ARM** templates.
- Developed and designed **AzureDevOps** pipeline to manage the resources across multiple subscriptions in Azure.
- Performed installation, configuration, upgrades, patching and maintenance of **Windows Server 2012/2016/2019**.
- **Configured and managed TCP/IP, DHCP, DNS, and Active Directory.**
- Created and managed **user accounts, permissions, and access rights**.
- Strong knowledge of installing, **configuring, tuning, maintaining, and monitoring** Windows Server **2008/2012 Enterprise server**.

- **Software Installation** and maintenances in **azure virtual machines**.
- Monitor system performance and **troubleshoot issues**.
- Ensure system security by setting up **firewall rules and antivirus software**.
- Managed **azure role-based** access control and **adding and removing** users from Active directory.
- **Provided configuration technical and administrative support** for Windows NT servers.
- **Software update management and storage system** implementation and maintenance.
- Created Azure infrastructure using **ARM templates, Azure pipelines** to build, test, and release, and managed **Azure Container Registry** to store private **Docker images**, which are deployed. Grafana Monitor to collect metrics and logs.
- Coordinate/assist developers with establishing and applying **appropriate branching, labeling/naming** conventions using **GIT** source control.
- Building/Maintaining **Docker container** clusters managed by **Kubernetes, Linux, Bash, GIT, Docker, on GCP**.
- Managed **host Kubernetes environment**, making it quick and easy to deploy and manage containerized applications without container orchestration expertise.
- Configured the **Kubernetes provider with Terraform** which is used to interact with resources supported by **Kubernetes** to create several services such as **Deployments, services, ingress rules, Config Map, secrets etc., in different Namespaces**.
- Experience in developing **APIs using Kubernetes** to manage and specify the copies of the **Docker containers** to run the actual servers in the **Azure cloud environment**. **Scheduled, deployed, and managed Container replicas onto a node cluster using Kubernetes**.
- Setting up **Kubernetes (k8s)** Clusters for running **microservices** and pushed **microservices** into production with Kubernetes backed Infrastructure. Development of automation of **Kubernetes clusters via playbooks in Ansible**.
- Implemented cluster services using **Docker and Kubernetes** to manage local deployments in Kubernetes by building a **self-hosted Kubernetes cluster** using **Terraform and Ansible** and deploying application containers.

Azure DevOps Engineer

Jan'2019 – Feb'2022

Client: Hiscox (Remote)

Responsibilities:

- Configured and deployed Azure Automation Scripts for a multitude of applications utilizing the Azure stack services including **Compute, Web & Mobile**, Blobs, **ADF**, Resource Groups, Azure Data Lake, HDInsight Clusters, Azure SQL and ARM focusing on automation, created quality gates in SonarQube dashboard and enforced in the pipelines to fail the builds when conditions not met.

- Creating and Managing Virtual Machines in Windows **Azure** and setting up communication with the help of Endpoints. Configured **Shared Access Signature (SAS)** tokens and Storage Access Policies in **Azure Cloud Infrastructure**.
- Deployed **Azure IaaS virtual machines (VMs)** and Cloud services (PaaS role instances) into secure **VNets** and subnets. Configured Azure Traffic Manager to build routing for user traffic
Infrastructure Migrations: Drive Operational efforts to migrate all legacy services to a fully Virtualized Infrastructure.
- Created **terraform** templates for provisioning virtual networks, subnets, VM Scale sets, Load balancers, and **NAT** rules and used terraform graph to visualize execution plan using the graph command. Configured **BGP** routes to enable ExpressRoute connections between on-premises data centers and **Azure** cloud.
- Experienced in migrating on-premises, **.Net** apps, DevOps platform to Azure CI/CD processes by utilizing **ARM templates**, **Azure DevOps**, Azure CLI, App services such as Repos, Pipelines, Web Apps, **PowerShell** and to monitor applications with Application Insights.
- Implemented **Ansible** to manage existing servers and automate the configuration of new servers also used Ansible Playbooks to setup Continuous Delivery Pipeline and Deployed micro services, including provisioning **Azure** environments using Ansible **Playbooks**.
- Created Azure infrastructure using **ARM templates**, **Azure pipelines** to build, test, and release, and managed **Azure Container Registry** to store private **Docker images**, which are deployed. Grafana Monitor to collect metrics and logs.
- Worked on creating dynamic **persistent volume** on **Azure Disk** and claimed this volume to a pod to make sure data is available even after killing the pod.
- Used **Azure Kubernetes** Service to deploy a managed Kubernetes cluster in Azure and created an **AKS cluster** in the Azure portal, with the Azure CLI, also used template driven deployment options such as Resource Manager templates and terraform.
- Implemented a **CI/CD** pipeline with **Docker**, **Jenkins**, and **GitHub** like when new GitHub branches are created Jenkins automatically attempts to build a new **Docker** container from it.
- Implemented a **Continuous delivery pipeline** with **Docker**, whenever a new GitHub branch gets started, our **Continuous Integration server** automatically attempts to build a new Docker container from it.
- Configured cluster operations in **Azure Kubernetes (AKS)** to deploy **microservices** with **CI/CD** system and created different namespaces service for each environment. Deployment in Kubernetes scale up pods, maintain services, load balancing, network policies, group **Docker** containers across different platform and managed entire package with **Helm charts**.
- Expertise in **Kubernetes** on premises for running/managing containers, images, and container snapshots, while creating a platform for automating, deploying, scaling of application containers across clusters of hosts.
- Performed Cloning, modifying, updating the **GIT** repos and push back to the **GitHub**. Imported and managed multiple corporate applications into GIT repositories and give access rights to the authorized developers.

- Implemented **Azure Boards** to track all issues pertaining to the software development lifecycle & integrated JIRA with repositories of GIT to track all code changes.

Role: DevOps Engineer

July'2018– Dec'2018

Client: Discover (Chicago, IL)

Responsibilities:

- Involved in three different developed teams and multiple simultaneous software releases.
- Involved in the Consulting, Analysis, Design, Development and Implementation of automated **Continuous Integration/Continuous Deployment** pipelines and test automation solutions.
- Coordinate/assist developers with establishing and applying appropriate branching, labeling/naming conventions using **GIT** source control.
- Integrated **maven** with **GIT** to manage and deploy project related tags.
- Installed and configured GIT and communicated with the repositories in **GITHUB**.
- Analyze and resolve conflicts related to merging of source code for GIT.
- Used **Jenkins** as continuous integration tool: creating new jobs, managing required plugins, configuring the jobs selecting required source code management tool, build trigger, build system, and post build actions, scheduled automatic builds, notifying the build reports.
- Implement, maintain, and support reliable, timely and reproducible builds for project teams.

ADB/ADF Developer / Azure Migration Engineer

Sept' 16 – July'18

Client: BCBS (Kansas, MO)

Responsibilities:

- Understanding the flow specifications and involved in development.
- Responsible for estimating the cluster size, monitoring, and troubleshooting of the **Spark data bricks cluster**.
- Created Notebooks using **Azure Data Bricks**.
- All the major **ETL/ELT (Transformations)** are done through **Azure Data Bricks**
- Developed Spark applications using **Scala** and **Spark-SQL** for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns.
- ADB Notebooks are invoked by the corresponding pipelines using **Azure Data Factory**.
- Involved in migrating data from on-prem to Azure using **Azure Data Factory**
- Created Pipelines in **ADF** using **Linked Services/Datasets/Pipeline/** to Extract, Transform, and load data from different sources like Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool and backwards.

- Developed **JSON Scripts** for deploying the Pipeline in **Azure Data Factory (ADF)** that process the data using the **SQL Activity**.
- Done necessary transformations on Un-Structured data through **Data Bricks**.
- Created **ADB Notebooks** for monthly and daily data to populate the proper data in Azure Data Warehouse.
- Involved in several stages of Data migration i.e., from **Universal Data Lake** to **Business Data Lake** and from **Business Data Lake** to **Data Warehouse**.
- Developed data warehouse using SSIS, SSAS tabular models.
- Identified the dimensions, fact tables and designed the data warehouse using **Star Schema**.
- Developed **Multi-Dimensional Objects (Cubes, Dimensions)** using **SQL Server Analysis Services (SSAS)**
- Created well designed dashboards using **Power BI**
- Migrate data from **traditional database systems** to **Azure databases**.
- Interacts with Business Analysts, Users, and SMEs on requirements.
- Recommend, Design and construct policies and standards that impact infrastructure operations and services but also improve overall business performance.
- Experience managing **Azure Data Lakes (ADLS)** and Data Lake Analytics and understanding of how to integrate with other Azure Services.
- Deploying Azure Resource Manager JSON Templates from PowerShell worked on Azure suite: Azure SQL Database, Azure Data Lake, Azure Data Factory, Azure SQL Data Warehouse, **AzureAnalysis** Service.